

Day 57 - Wireless Security

Wednesday, September 9, 2026

8:38 PM

Wireless Security Info

Security is very important in **wireless** networks, because **wireless** signals are not contained within a wire, so, any device within range can pick up these signals and receive the traffic. That's why we need to encrypt the traffic sent between the **wireless** devices and the AP.

- This is different to **wired** networks where traffic is only encrypted when sent over untrusted networks, such as the Internet.

Authentication

All clients must be **authenticated** before they can associate with an AP.

- In a corporate setting, only trusted users/devices should be given access to the network.
 - A separate **SSID** which doesn't have access to the corporate network can be provided for guest users.
- Ideally, clients should also **authenticate** the **AP** to avoid associating with a malicious **AP**.
- There are multiple ways to **authenticate**:
 1. **Password**
 2. **Username/Password**
 3. **Certificates**

Encryption

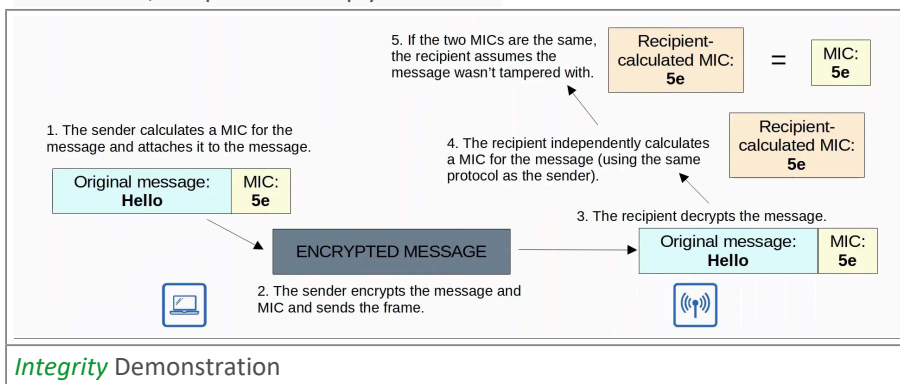
Traffic sent between clients and APs should be encrypted so that it can't be read by anyone else except the AP and the client.

- All devices on the **WLAN** will use the same protocol, however, each client will use a unique **encryption/decryption** key so that other devices can't read its traffic.
- A '**group key**' is used by the **AP** to **encrypt** traffic that it wants to send to all of its clients.
 - All of the clients associated with the **AP** keep that key so they can encrypt the traffic.

Integrity

Integrity ensures that a message is not modified by a third-party.

- The message that is sent by the **source** host should be the same as the message that is received by the **destination** host.
- A **Message Integrity Check (MIC)** is added to messages to help protect their **integrity**. This is done through a few steps:
 1. The **sender** calculates a **MIC** for the message and attaches it to the message.
 2. The **sender** encrypts the message and **MIC** and sends the frame.
 3. The **recipient** decrypts the message.
 4. The **recipient** independently calculates a **MIC** for the message (using the same protocol as the **sender**).
 5. If the two **MICs** are the same, the **recipient** assumes the message wasn't tampered with.



Authentication Methods

There are many *authentication* methods available that provide devices access with the network. Some of them more secure than others. They include:

1. *Open Authentication*
2. *WEP (Wired Equivalent Privacy)*
3. *EAP (Extensible Authentication Protocol)*
4. *LEAP (Lightweight EAP)*
5. *EAP-FAST (EAP Flexible Authentication via Secure Tunneling)*
6. *PEAP (Protected EAP)*
7. *EAP-TLS (EAP Transport Layer Security)*

1. Open Authentication

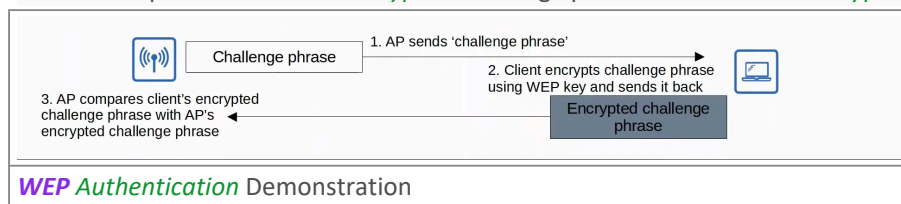
Included in the original 802.11 standard.

- The client sends an *authentication* request, and the **AP** accepts it.
- Not a secure *authentication* method.
- After the client is *authenticated* and *associated* with the **AP**, it's possible to require the user to *authenticate* via other methods before access is granted to the network.

2. WEP (Wired Equivalent Privacy)

WEP is used to provide both *authentication* and *encryption* of wireless traffic.

- For *encryption*, *WEP* uses the **RC4** algorithm.
- *WEP encryption* is not secure and can be easily cracked.
- *WEP* is a 'shared-key' protocol, requiring the *sender* and the *receiver* to have the same key.
- *WEP* keys can be **40 bits** or **104 bits** in length.
 - These keys are combined with a **24 bit 'Initialization Vector (IV)'** to bring the total length to **64 bits** or **128 bits**.
- *WEP authentication* can be done through the following steps:
 1. The **AP** sends a 'challenge phrase'.
 2. The client *encrypts* the challenge phrase and using *WEP* keys and sends it back.
 3. The **AP** compares the client's *encrypted* challenge phrase with the **AP's encrypted** challenge phrase.



3. EAP (Extensible Authentication Protocol)

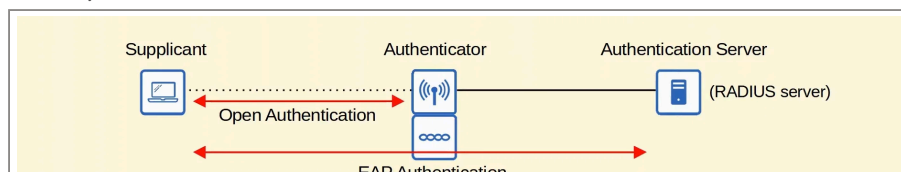
EAP is an *authentication* framework. It defines a standard set of *authentication* functions that are used by the various *EAP Methods*

- There are four *EAP Methods*: *LEAP*, *EAP-FAST*, *PEAP*, and *EAP-TLS*
- *EAP* is integrated with **802.1X**, which provides port-based network access control.

802.1X Explanation

802.1X is used to limit network access for clients connected to a LAN or WLAN until they *authenticate*.

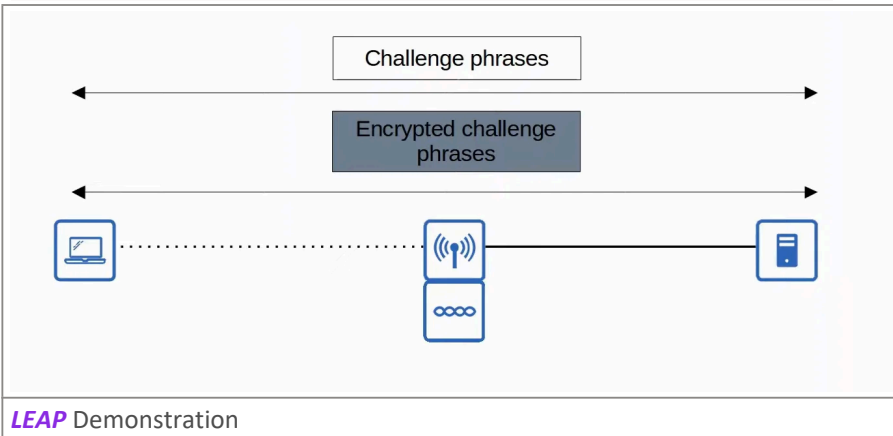
1. *Supplicant*
 - The device that wants to connect to the network.
2. *Authenticator*
 - The device that provides access to the network.
3. *Authentication Server (AS)*
 - The device that receives client credentials and permits/denies access.
 - Usually a **RADIUS** server.



4. LEAP (Lightweight EAP)

LEAP was developed by Cisco as an improvement over **WEAP**.

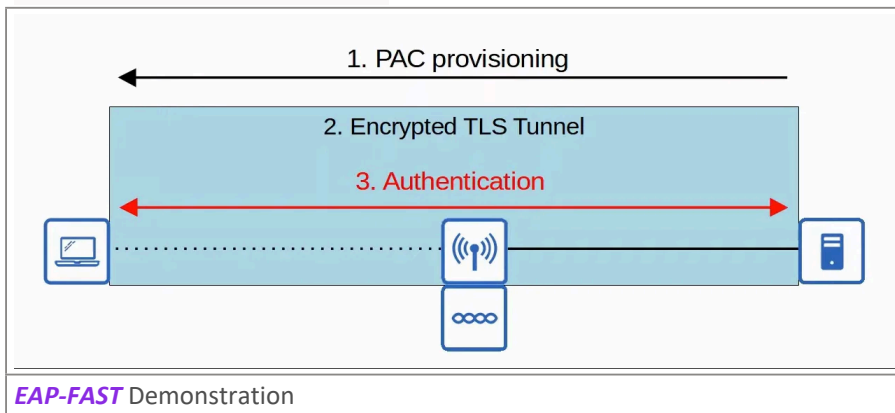
- Clients must provide a **username** and **password** to **authenticate**.
- In addition, **mutual authentication** is provided by both the client and server sending a challenge phrase to each other.
- **Dynamic WEAP** keys are used, meaning that the **WEAP** keys are changed frequently.
- Like **WEAP**, **LEAP** is considered vulnerable and should not be used anymore.



5. EAP-FAST (EAP Flexible Authentication via Secure Tunneling)

EAP-FAST was also developed by Cisco

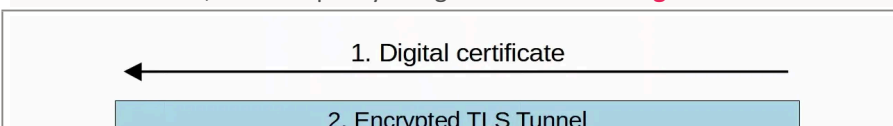
- Consists of three phrases:
 1. A **Protected Access Credentials (PAC)** is generated and passed from the server to the client.
 2. A secure **TLS** tunnel is established between the client and **authentication** server.
 3. Inside of the secure (**encrypted**) **TLS** tunnel, the client and server communicate further and **authenticate/authorize** the client.

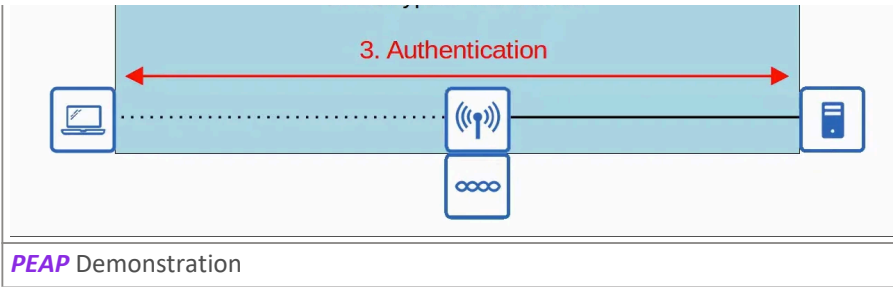


6. PEAP (Protected EAP)

Similarly to **EAP-FAST**, **PEAP** involves establishing a secure TLS tunnel between the client and server.

- Instead of a **PAC**, the server has a digital certificate.
- The client uses a digital certificate to **authenticate** the server.
- The certificate is also used to establish a **TLS** tunnel.
- Because only the server provides a certificate for **authentication**, the client must still be **authenticated** within the secure **TLS** tunnel, for example by using **Microsoft Challenge-Handshake Authentication Protocol (MS-CHAP)**

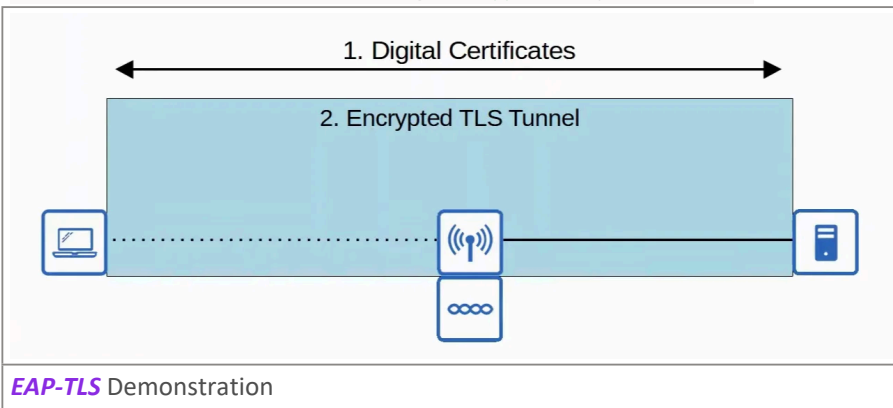




7. EAP-TLS (EAP Transport Layer Security)

EAP-TLS is the most secure **EAP Method**, however, it is the hardest one to implement.

- Whereas **PEAP** only requires the **Authentication Server (AS)** to have a certificate, **EAP-TLS** requires a certificate on every single client.
- Because the client and server **authenticate** each other with digital certificates, there is no need to **authenticate** the client within the **TLS** tunnel.
- The **TLS** tunnel is still used to exchange **encryption** key information.



Encryption and Integrity Methods

There are three main **encryption** and **integrity** methods you need to know for the CCNA.

- TKIP (Temporal Key Integrity Protocol)**
- CCMP (Counter/CBC-MAC Protocol)**
- GCMP (Galois/Counter Mode Protocol)**

1. TKIP (Temporal Key Integrity Protocol)

WEP was found to be vulnerable, but **wireless** hardware at the time was built to use **WEP**.

- TKIP** was created as a temporary solution until a new standard was created and new hardware was built.
- It is used in **WPA** version **1** (aka. **WPA**).
- TKIP** adds various security features, including:
 - A **MIC** to protect **integrity**.
 - The **MIC** includes the **sender** MAC address.
 - Timestamp has been added to the **MIC**.
 - Key mixing algorithm** to create unique **WEP** keys for each frame.
 - Doubled the size of the **Initialization Vector (IV)**.
 - A **TKIP sequence number** is used to keep track of frames sent from each **source** MAC address.

2. CCMP (Counter/CBC-MAC Protocol)

CCMP was developed after TKIP and is more secure.

- It is used in **WPA** version **2** (aka. **WPA2**).
- To use **CCMP**, it must be supported by the device's hardware.
- CCMP** consists of two different algorithms to provide **encryption** and **MIC**:
 - Advanced Encryption Standard (AES)** counter mode
 - AES** is the most secure **encryption** protocol currently available.
 - There are multiple modes of operation for **AES**. **CCMP** uses 'counter mode'.

2. Cipher Block Chaining Message Authentication Code (CBC-MAC)

- Used as a **MIC** to ensure the *integrity* of messages.

3. GCMP (Galois/Counter Mode Protocol)

GCMP is more secure and efficient than CCMP.

- Its increased efficiency allows for higher data throughput than **CCMP**.
- It is used in **WPA** version **3** (aka. **WPA3**).
- **GCMP** consists of two different algorithms to provide *encryption* and **MIC**:
 1. **Advanced Encryption Standard (AES)** counter mode
 2. **Galois Message Authentication Code (GMAC)**
 - Used as a **MIC** to ensure the *integrity* of messages.

Summary

1. **TKIP (Temporal Key Integrity Protocol)**
 - Based on **WEP**, but more secure.
 - Should not be used in modern networks.
 - Used in **WPA**
2. **CCMP (Counter/CBC-MAC Protocol)**
 - **AES** counter mode for *encryption*
 - **CBC-MAC** for **MIC**
 - Used in **WPA2**
3. **GCMP (Galois/Counter Mode Protocol)**
 - **AES** counter mode for *encryption*
 - **GMAC** for **MIC**
 - Used in **WPA3**

Wi-Fi Protected Access (WPA)

The Wi-Fi alliance has developed three WPA certifications for wireless devices

1. **WPA**
 2. **WPA2**
 3. **WPA3**
- All of the above **WPA** versions support two *authentication* modes:
 1. **Personal Mode**
 - A **pre-shared key (PSK)** is used for *authentication*. When you connect to a home Wi-Fi network, enter the password and are *authenticated*, that is **personal** mode.
 - Common for small networks.
 - The **PSK** itself is not sent over the air. A four-way handshake is used for *authentication*, and the **PSK** is used to generate *encryption* keys.
 2. **Enterprise Mode**
 - **802.1X** is used with an **Authentication Server (AS)** (**RADIUS** server).
 - No specific **EAP method** is specified, so all are supported.

1. WPA

The WPA certification was developed after **WEP** was proven to be vulnerable and includes the following protocols:

- **TKIP** (based on **WEP**) provides *encryption*/**MIC**.
- **802.1X** authentication (**Enterprise mode**) OR **PSK** (**Personal mode**)

2. WPA2

WPA2 was released after WPA and includes the following protocols:

- **CCMP** provides *encryption*/**MIC**.
- **802.1X** authentication (**Enterprise mode**) OR **PSK** (**Personal mode**)

3. WPA3

WPA3 was released in 2018 and includes the following protocols:

- **GCMP** provides *encryption*/**MIC**.
- **802.1X** authentication (**Enterprise mode**) OR **PSK** (**Personal mode**)
- **WPA3** also provides several additional security features, for example:

- **PMF** (*Protected Management Frames*)
 - Protecting **802.11** management frames from eavesdropping/forging.
- **SAE** (*Simultaneous Authentication of Equals*)
 - Protects the four-way handshake when using *personal mode authentication*.
- **Forward Secrecy**
 - Prevents data from being decrypted after it has been transmitted over the air. So, an attacker can't capture wireless frames and then try to decrypt them later.